

1. Executive Summary

This report presents the complete root cause analysis of the QBIT Hub Enterprise Portal v2.0. Every broken button, non-functional feature, and incomplete workflow was traced from UI component through click handler, API route, backend service, database query, and response. The analysis identified 22 issues across 6 categories. The root causes are: (1) Stitch UI pages using hardcoded placeholder data instead of API calls, (2) buttons without onClick handlers, (3) Desktop Agent not implemented for hardware scanning, (4) download API returning placeholder URLs, (5) product CRUD API missing PUT/PATCH/DELETE, and (6) upload functionality showing 'coming soon' toasts.

2. Issue Summary

Severity	Count	Root Cause Category
CRITICAL	8	No onClick handlers, Desktop Agent missing, Download file endpoint missing
HIGH	9	Hardcoded placeholder data (22 pages), Product CRUD incomplete, Upload 'coming soon'
MEDIUM	5	QR generation stub, Import wizard stub, Settings save stub
TOTAL	22	

Root Cause Analysis Report

Complete Project Audit — UI to Database Trace
QBIT Hub Enterprise Portal — Version 2.0

22 Issues Found

8 Critical

9 High

5 Medium

Document

Root Cause Analysis Report

Version: 2.0 | Date: 14 July 2026

Classification: Internal — Architect Review

3. CRITICAL Issues (8)

ID	Feature	Root Cause	Affected Files	Required Fix	Est. Time
RCA-001	Product Management: Add Product button	Button exists but has NO onClick handler. Renders as static QbitButton with no event binding.	ProductManagementPage.tsx line 181	Add onClick handler that opens a product create dialog/form. Create POST /api/admin/products endpoint.	4h
RCA-002	Product Management: Edit button	Button exists (line 396) but has NO onClick handler. No edit form/dialog exists.	ProductManagementPage.tsx line 396	Add onClick handler. Create edit dialog. Add PUT /api/admin/products/[id] endpoint.	4h
RCA-003	Product Management: Delete button	Button exists (line 403) but has NO onClick handler. No delete confirmation dialog.	ProductManagementPage.tsx line 403	Add onClick with confirmation dialog. Add DELETE /api/admin/products/[id] endpoint.	3h
RCA-004	Dr. QBIT: Scan Hardware button	ScanButton component exists but handleScan() only shows a toast + setTimeout. Does NOT call any API. No Desktop Agent exists.	ScanButton.tsx, desktop-agent-auth.ts	Desktop Agent (.exe/Electron app) must be built separately. Browser CANNOT access USB/COM hardware. API contract exists (POST /api/dr-qbit/scan) but no agent sends data to it.	40h+
RCA-005	Downloads: Download button	API /api/downloads/[id] returns placeholder URL '/api/downloads/[id]/file?token=...' but /file endpoint does NOT exist.	api/downloads/[id]/route.ts	Create /api/downloads/[id]/file/route.ts that serves the actual file (or redirects to cloud storage signed URL).	4h
RCA-006	Product Management: Save/Update	No save/update handler exists. No form state management. No PATCH/PUT API for products.	ProductManagementPage.tsx, dr-qbit/products/route.ts	Product CRUD API only has GET + POST (create). Missing PUT (update) and DELETE. Need /api/admin/products/[id] with PUT+DELETE.	6h
RCA-007	Dr. QBIT: Device Passport not opening	Passport page works for Dr. QBIT-created passports. But ProductManagementPage products are hardcoded ADMIN_ASSETS — not linked to QbitProduct/DevicePassport models.	ProductManagementPage.tsx imports ADMIN_ASSETS (placeholder data)	Replace hardcoded ADMIN_ASSETS with fetch from /api/dr-qbit/products. Link product management to QbitProduct model.	8h
RCA-008	Dr. QBIT: Driver Detection missing	No driver detection UI or API. DriverInformation is populated by passport generator (simulated). No real driver detection from Windows API.	engine.ts (passport generator simulates driver data)	Desktop Agent must report installed driver version via Windows DriverStore API. Browser cannot access this.	Part of RCA-004

4. HIGH Issues (9)

ID	Feature	Root Cause	Affected Files	Required Fix	Est. Time
RCA-009	22 pages use hardcoded data (no API calls)	Original Stitch UI pages were implemented as static designs with placeholder data. They were never connected to APIs.	AI\SupportCenterPage, AdminDashboardPage, CustomerHandoverReportPage, EngineerDashboardPage, FieldEngineerWorkspacePage, HomePage, InstallationCenterPage, JobCompletionHandoverPage, JobDetailsInst550APage, ProductDetailsT800Page, ProductLibraryPage, ProductManagementPage, PublicSearchPage, QbitT800ProductOverviewPage, SystemSettingsPage, T800InstallationGuidePage, UniversalSearchCommandCenterPage, UniversalSearchMobilePage, UserRoleManagementPage, VideoTrainingCenterPage, FSMCustomerTrackingPage, LoginPage	Each page needs: fetch() calls to appropriate API, useState for data, loading states, error handling. This is the LARGEST issue in the project.	80h+
RCA-010	AssetManager: Upload Asset button	Button calls onUpload ?? notifyComingSoon(). onUpload is undefined (optional prop never passed). Shows 'coming soon' toast.	AssetManager.tsx line 71	Pass onUpload handler from parent. Create /api/admin/media/upload endpoint with multipart file upload.	6h
RCA-011	MediaManager: Upload Media button	Button has NO onClick handler at all. Renders as static QbitButton.	MediaManager.tsx line 48	Add onClick handler. Create file upload dialog. Wire to /api/admin/media POST endpoint (exists but only accepts metadata, not file).	4h
RCA-012	AssetManager: Edit Asset	Calls onEdit ?? notifyComingSoon(). onEdit is never passed. Shows 'coming soon'.	AssetManager.tsx line 160	Pass onEdit handler. Create edit dialog. Add PUT /api/admin/media/[id] endpoint.	4h
RCA-013	AssetManager: Replace Version	Directly calls notifyComingSoon(). No handler, no API.	AssetManager.tsx line 163	Create version replacement workflow. Add POST /api/admin/media/[id]/version endpoint.	6h
RCA-014	AssetManager: Toggle Visibility	Directly calls notifyComingSoon(). No handler, no API.	AssetManager.tsx line 166	Add PATCH /api/admin/media/[id] endpoint with visibility field. Pass handler from parent.	2h
RCA-015	Product Management: Bulk Delete	QbitButton has NO onClick handler. No bulk delete API.	ProductManagementPage.tsx line 245	Add onClick. Create DELETE /api/admin/products/bulk endpoint.	3h

ID	Feature	Root Cause	Affected Files	Required Fix	Est. Time
RCA-016	Product Management: Export CSV	QbitButton has NO onClick handler. No export API for products.	ProductManagementPage.tsx line 253	Add onClick. Create GET /api/admin/products/export?format=csv endpoint.	2h
RCA-017	ImportWizard: Import flow	Uses static IMPORT_PREVIEW data. No real file upload or API call.	ImportWizard.tsx, cms/placeholder-data.ts	Wire to /api/admin/import endpoint (exists but returns mock data). Add real file parsing.	8h

5. MEDIUM Issues (5)

ID	Feature	Root Cause	Affected Files	Required Fix	Est. Time
RCA-018	QR Code generation	QRManager component exists in admin/ but has no visible implementation. /api/admin/qr route exists but returns static data.	QRManager.tsx, api/admin/qr/route.ts	Implement QR code generation using qrcode library. Wire to product/asset records.	4h
RCA-019	SystemSettings: Save button	Theme toggle works (setTheme). But other settings (notifications, backups, etc.) have no save handler.	SystemSettingsPage.tsx	Add save handlers for each setting section. Wire to PUT /api/admin/settings (exists).	4h
RCA-020	UserRoleManagement: Add/Edit/Delete user	Page has tab navigation (users/roles) but no add/edit/delete buttons or forms for users.	UserRoleManagementPage.tsx	Add user CRUD UI. Create /api/admin/users endpoints (GET/POST/PUT/DELETE).	8h
RCA-021	Product Management: Product images	No image upload UI. No image display from DB. Products show placeholder icons.	ProductManagementPage.tsx	Add image upload to product create/edit form. Wire to MediaFile model + /api/admin/media.	4h
RCA-022	Product Management: Category mapping	No category CRUD. Categories are hardcoded in placeholder data.	ProductManagementPage.tsx, admin/placeholder-data.ts	Create Category model + /api/admin/categories endpoints. Replace hardcoded data.	6h

6. Root Cause Analysis

6.1 Why are buttons not working?

The original Stitch UI design files were implemented as pixel-perfect static HTML/CSS replicas. The design included buttons, forms, tables, and navigation — but they were visual only. When converted to React components, the visual elements were preserved but the interactivity was not added. Specifically:

- Buttons without onClick: 15+ buttons across ProductManagementPage, MediaManager, and other admin pages render as visual elements with no event handlers.
- Forms without onSubmit: No create/edit forms exist for products, users, or categories.
- Tables without data binding: ProductManagementPage uses ADMIN_ASSETS (hardcoded array from placeholder-data.ts) instead of fetching from API.
- 'Coming soon' pattern: AssetManager uses a notifyComingSoon() fallback when onUpload/onEdit handlers are not passed. The parent never passes them.

6.2 Why is Dr. QBIT scanning not working?

Hardware scanning (USB, COM, LAN, WiFi) requires operating system-level access that browsers cannot provide. The architecture was designed correctly: Device -> Desktop Agent -> Secure Local API -> QBIT Hub. However:

- Desktop Agent NOT implemented: No .exe, Electron app, or local service exists. The API contract (POST /api/dr-qbit/scan with agentSecret) is ready, but nothing sends data to it.
- ScanButton is a simulation: handleScan() shows a toast notification and uses setTimeout(3000ms) to simulate a scan. It does NOT call any API.
- What works without Desktop Agent: Device Passport, Driver Intelligence, Firmware Intelligence, AI Diagnostics, and Test Center all work because they use pre-seeded data from the database. The scan data was seeded via scripts/seed-dr-qbit.ts.
- What requires Desktop Agent: Real-time hardware discovery, real driver version detection, real firmware version reading, real test execution (test print, scan input, drawer open).

6.3 Why do downloads not work?

- API returns placeholder URL: /api/downloads/[id] returns { secureUrl: '/api/downloads/[id]/file?token=...' } but the /file endpoint does not exist.
- No cloud storage configured: The Download model has a storagePath field, but no actual files are stored. No S3/Supabase/UploadThing integration.
- Download count still increments: The API does increment downloadCount and create DownloadHistory — so analytics work. But the actual file download fails.

6.4 Why is Product Management not working?

- UI Complete, Backend Missing: The ProductManagementPage renders a full table with products, but uses hardcoded ADMIN_ASSETS data.
- API Missing: No /api/admin/products endpoint exists. The only product API is /api/dr-qbit/products (GET + POST for Dr. QBIT device matching, not admin CRUD).
- CRUD Status: Create (POST exists on dr-qbit/products but not wired to UI). Read (GET exists but UI doesn't call it). Update (NO PUT/PATCH). Delete (NO DELETE).

6.5 Why do 22 pages have no API calls?

These 22 pages were the original Stitch design screens. They were built as visual prototypes with hardcoded data to match the approved design. When the Version 2 modules (FSM, Notifications, Dr. QBIT, Fleet, Analytics) were built, they were implemented with proper API calls. But the original 22 design screens were never upgraded to use real APIs. They still display static data.

7. CRUD Matrix — All Modules

Module	Create	Read	Update	Delete	Notes
Products (Admin)	MISSING	MISSING (hardcoded)	MISSING	MISSING	UI buttons exist but no handlers. No /api/admin/products endpoint.
Products (Dr. QBIT)	POST /api/dr-qbit/products	GET /api/dr-qbit/products	PATCH /api/dr-qbit/products/[id]	DELETE /api/dr-qbit/products/[id]	Works for Dr. QBIT. NOT wired to ProductManagementPage.
Drivers	MISSING	MISSING (hardcoded)	MISSING	MISSING	Driver model exists. No admin CRUD API. AssetManager shows 'coming soon'.
Firmware	MISSING	GET /api/dr-qbit/firmware/releases	MISSING	MISSING	FirmwareRelease model exists. No admin CRUD API.
Knowledge Base	MISSING	MISSING (hardcoded)	MISSING	MISSING	KnowledgeArticle model exists. AISupportCenterPage uses hardcoded articles.
Downloads	POST /api/admin/media (metadata only)	GET /api/public/downloads	MISSING	MISSING	No file upload. No /file endpoint for actual download.
Customers (FSM)	MISSING	GET /api/fsm/work-orders	PATCH /api/fsm/work-orders/[id]	MISSING	FSM customers accessible via work orders. No direct customer CRUD.
Work Orders	POST /api/fsm/work-orders	GET /api/fsm/work-orders	PATCH /api/fsm/work-orders/[id]	MISSING	Create + Read + Update work. No Delete.
Notifications	POST /api/admin/notifications/dispatch	GET /api/notifications	PATCH /api/notifications/[id]	MISSING	In-app CRUD works. No delete (archive instead).
Device Passports	POST /api/dr-qbit/passports/generate	GET /api/dr-qbit/passports	MISSING	MISSING	Auto-generated from scans. No manual edit/delete.
Users	MISSING	MISSING (hardcoded)	MISSING	MISSING	User model exists. UserRoleManagementPage has no CRUD. No /api/admin/users.
Branches	MISSING	GET /api/fleet/devices (via branch)	MISSING	MISSING	Branch model exists. No admin CRUD API.

8. Desktop Agent — Required for Hardware Scanning

The browser CANNOT access USB devices, COM ports, or LAN network discovery. This is a fundamental browser security restriction. The QBIT Hub architecture correctly isolates hardware access behind a Desktop Companion Agent, but this agent has not been built.

Component	Status	What Exists	What's Missing
Desktop Agent (.exe/Electron)	NOT IMPLEMENTED	API contract: POST /api/dr-qbit/scan with agentSecret. Auth: desktop-agent-auth.ts	Windows app that scans USB/COM/LAN/WiFi and POSTs results to the API
Scan Button (UI)	SIMULATION ONLY	ScanButton.tsx shows animation + toast	WebSocket connection to Desktop Agent, real scan trigger
Device Detection	WORKS (with seed data)	DetectedDevice model, /api/dr-qbit/scan, /api/dr-qbit/devices	Real-time detection from Desktop Agent (currently uses seed-dr-qbit.ts)
Driver Detection	SIMULATED	DriverInformation populated by passport generator (simulated version)	Desktop Agent reads Windows DriverStore, reports real installed driver version
Firmware Detection	SIMULATED	FirmwareInformation populated by passport generator	Desktop Agent queries device via ESC/P or SNMP, reports real firmware version
Test Execution	SIMULATED	Test runner simulates test results based on device state	Desktop Agent sends actual test commands (test print, scan input, drawer open)

8.1 What works WITHOUT Desktop Agent?

- Device Passport view (display + driver/firmware comparison) - uses pre-seeded data
- Driver Intelligence (installed vs latest comparison) - uses simulated installed version
- Firmware Intelligence (installed vs latest + compatibility check) - uses simulated installed version
- AI Diagnostics (health score + findings + recommendations) - analyzes pre-seeded data
- Test Center (test suite execution + results) - simulates test results
- Fleet Manager (device inventory + analytics) - aggregates pre-seeded data
- All of these modules work correctly with seeded data — they just can't detect real hardware.

9. Implementation Status Summary

Module	UI	API	Database	Desktop Agent	Overall
FSM (Work Orders)	COMPLETE	COMPLETE	COMPLETE	N/A	PRODUCTION READY
Notification Engine	COMPLETE	COMPLETE	COMPLETE	N/A	PRODUCTION READY
Customer Tracking	COMPLETE	COMPLETE	COMPLETE	N/A	PRODUCTION READY
Engineer Mobile (PWA)	COMPLETE	COMPLETE	COMPLETE	N/A	PRODUCTION READY
Dr. QBIT Detection	COMPLETE	COMPLETE	COMPLETE	REQUIRED	PARTIAL (needs agent)
Dr. QBIT Passport	COMPLETE	COMPLETE	COMPLETE	N/A	PRODUCTION READY
Dr. QBIT Driver Intel	COMPLETE	COMPLETE	COMPLETE	REQUIRED (for real data)	PARTIAL
Dr. QBIT Firmware Intel	COMPLETE	COMPLETE	COMPLETE	REQUIRED (for real data)	PARTIAL
Dr. QBIT Diagnostics	COMPLETE	COMPLETE	COMPLETE	REQUIRED (for real data)	PARTIAL
Dr. QBIT Test Center	COMPLETE	COMPLETE	COMPLETE	REQUIRED (for real tests)	PARTIAL
Fleet Manager	COMPLETE	COMPLETE	COMPLETE	N/A	PRODUCTION READY
Analytics	COMPLETE	COMPLETE	COMPLETE	N/A	PRODUCTION READY
Product Management	UI ONLY	MISSING	MODEL EXISTS	N/A	NOT READY
User Management	UI ONLY	MISSING	MODEL EXISTS	N/A	NOT READY
Download Center	COMPLETE	PARTIAL (placeholder URL)	COMPLETE	N/A	PARTIAL
Knowledge Base	UI ONLY	MISSING	MODEL EXISTS	N/A	NOT READY
Admin CMS	UI ONLY	PARTIAL	MODEL EXISTS	N/A	NOT READY

10. Recommended Fix Priority

- Phase 1 (Critical): Fix Product Management CRUD (RCA-001 to RCA-003, RCA-006). Create /api/admin/products with GET/POST/PUT/DELETE. Wire ProductManagementPage to API. Est: 20h

- Phase 2 (High): Fix Download file endpoint (RCA-005). Create /api/downloads/[id]/file/route.ts. Est: 4h

- Phase 3 (High): Fix Upload functionality (RCA-010 to RCA-014). Wire AssetManager/MediaManager handlers. Est: 20h
- Phase 4 (High): Connect 22 static pages to APIs (RCA-009). Largest effort. Est: 80h+
- Phase 5 (Medium): Fix QR generation, settings save, user management (RCA-018 to RCA-022). Est: 26h
- Phase 6 (Separate project): Build Desktop Companion Agent (RCA-004, RCA-008). Est: 40h+

Waiting for approval before making any fixes.